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Planning of carbon capture and storage with pinch analysis techniques

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ABSTRACT

Carbon capture and storage (CCS) is a means for reducing carbon dioxide (CO_2) emissions from fossil fuel combustion in power generation and industrial processes. It involves the capture of CO_2 for subsequent storage in various geological formations. The selection and matching of the power plants and storage sites are often an issue of optimisation due to various constraints, i.e., time of availability, injection rate, and storage capacity limits. In this work, a novel graphical targeting tool based on pinch analysis is proposed to address the planning problem of the storage of captured CO_2 from power generating plants into corresponding reservoirs. The main consideration for the problem is the time of availability of the latter, since reservoirs need to be developed prior to CO_2 storage. The time limitation is addressed by the graphical technique where time is taken as the governing element in solving the problem. Hypothetical examples are used to elucidate the proposed approach.

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Keywords: Carbon capture and storage; CO2 emission reduction; Process integration; Pinch analysis; Targeting; Source-sink matching

1. Introduction

Recently, more countries around the world have committed to progressive carbon dioxide (CO_2) emission reduction over time, as the world struggles to balance the growth in energy requirement and environment conservation for a sustainable future. This trend has contributed to the growing interest in carbon capture and storage (CCS) initiatives which have been widely discussed as means to cut emission from processes involving fossil fuel combustion. CCS mainly involves capturing CO_2 from large point sources such as fossil fired power plants, and subsequently storing the captured CO_2 in geological reservoirs to ensure that it does not enter the atmosphere. Large-scale deployment of CCS can reduce the total carbon emission levels of a country or region, even with an increase in power generation capacity and demand. The emission level from retrofitted power plants will be lower than a modern conventional plant (without CCS) using the same fuel by approximately 80–90%, based on data from Intergovernmental Panel on Climate Change (Metz et al., 2005).

Promising carbon capture (CC) technologies include oxy-fuel combustion, which involves burning fuel in the absence of nitrogen to yield CO_2-rich flue gas (Wall et al., 2009). Besides, other technologies developed for CC includes chemical looping combustion, pre-combustion capture (using integrated gasification combined cycles, IGCC), or post combustion capture (via flue gas scrubbing with solvents such as amines) (Yang et al., 2008). Retrofitting existing power plants with CCS allows for continued used of fossil fuel such as coal and natural gas while drastically reducing emission. Besides, CC technologies will be able to cushion the pressure imposed on the development of large scale application in renewable energy to replace fossil fuels in the future. Total replacement of fossil fuels for power generation is not expected to materialise in near future as dependency on fossil fuel is expected to continue and expand. The World Energy Outlook (WEO)

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2009 claims that fossil fuels remain the dominant sources of primary energy accounting for more than three quarters of overall increase in energy use between 2007 and 2030 (IEA, 2009a). As such, CCS can be used in conjunction with other "technology wedges" such as renewables to reduce greenhouse gas emissions from energy use (Williams et al., 2012). The main disadvantage of using CC in power plants is the reduction of power output due to the energy requirements of capturing CO_2; this power loss, combined with capital outlays, can result in an increase of power generation costs by 40–60% (IEA, 2009b).

Captured CO_2 from the power plants may be either stored in reservoirs such as unused saline aquifers (Bachu and Adams, 2003), inaccessible coal seams and geological reservoirs or transported to depleted oil production fields for enhanced oil recovery (EOR). The captured CO_2 can be stored permanently in geological reservoirs, where deep aquifers can contain large amount of CO_2 in the form of compressed gas, liquid or aqueous solution under high formation pressure. Koide et al. (1999) conducted a study on utilising the unused reservoir for permanent CO_2 storage and concluded that from the point of storage capacity and economics, CO_2 containment in geological reservoir is a feasible solution. In the context of EOR, CO_2 can be used as injection gas into oil producing wells to increase the amount of oil recoverable from an oil field (US Department of Energy, 2012). CO_2 injection into oil wells can be considered as a viable option to extend the production life of the oil fields and at the same time contribute to reduction in CO_2 emission to the atmosphere. However, in this case both injection rate and period of operation for CO_2 capture are constrained by oil extraction considerations.

Although the selection and utilisation of CO_2 for EOR looks to be the more promising method for CO_2 storage from an economic viewpoint compared to permanent reservoir storages, the amount required for CO_2 storage may be limited and there are regional disparities in the occurrence of oil and gas fields. Required CO_2 injection may be subjected to oil production forecast and capability of the oil and gas production facilities to handle the increase in oil recovery from the existing field. In addition, the timing and availability of CO_2 sinks and sources, coupled with injection facilities does require comprehensive planning to optimise the sink–source matching. Turk et al. (1987) presented pure integer linear programming model in an early attempt to optimise the matching of CO_2 sink–sources for EOR.

Among the factors considered during feasibility planning stage in matching of CO_2 sinks and sources is the time availability, CO_2 storage capacity and injection rate. In addition, grid-related aspects such as energy losses need to be accounted for. Linear programme (LP) and mixed integer linear programme (MILP) models were developed by Foo et al. (2010a), Pękala et al. (2010) and Tan et al. (2010), where CCS retrofit was performed for carbon-constrained energy planning, taking into account the need to compensate for power losses due to CC. Tan et al. (2012, 2013) and Lee and Chen (2012) later extended the MILP model into continuous-time and discrete-time variants for the matching of CO_2 sources and sinks under temporal, injection rate and storage capacity constraints.

In this paper, the CCS planning problem is addressed via a novel graphical tool, which is extended from the human resource management problem based on pinch analysis technique (Foo et al., 2010b). However, a new definition for resource sinks and sources is used here. In this work, the availability of CO_2 storage reservoirs is defined as *storage sources* that provides "service" to CO_2 emitting power plants, which takes the analogy of man power availability in completing a given task in human resource management problem (Foo et al., 2010a, 2010b). In effect, storage capacity is treated as a resource that is available in geological reservoirs, and is consumed by the power plants that generate captured CO_2 streams. These plants thus act as *storage demands* (analogy to task to be completed in human resource management problem – Foo et al., 2010a, 2010b), are to be matched to the appropriate storage source. This new definition for the role and contribution of the sink–source is shown in Fig. 1. Note that the convention used in this work where physical sources of CO_2 emission (i.e., power plants) are taken as demand for the *service* of CO_2 disposal, and conversely the storage reservoirs which act as physical sinks are taken as sources of this service. Hence, the disposal service flows in the opposite direction as the captured CO_2 itself. This convention was proposed in the context of life cycle assessment by Tan et al. (2008); its logical basis in the context of general waste treatment and disposal systems is discussed in detail by Heijungs and Suh (2002). From the above description, the sink–source definition is different than the assumption made in the work of Tan et al. (2012, 2013) where power plants are considered as CO_2 sources while the storage as sinks.

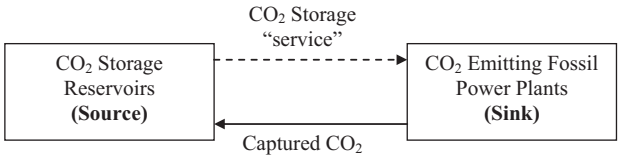


Fig. 1 – CO_2 source and demand convention for CCS planning problem (Tan et al., 2008).

The paper will first approach the problem using macroscopic approach based on overall matching of storage source available and storage demand. The timing of the requirement for the additional storage resource will be optimised using targeting methods to ensure that additional storage is only introduced when required. This will provide better estimation of the urgency of the need for a new reservoir to be developed to support the deficit of storage capacity in the CCS network.

Subsequently, a detailed approach will be used to identify the pairing options available for a particular power plant to a specific storage site. The various options identified will be developed into roadmaps for further evaluation on project economics and logistics for the best option to proceed with engineering and construction (however this is outside the scope of this work).

2. Problem statement

The formal statement derived for the CCS planning problem is given as follows:

- The CCS network is assumed to be comprised of i storage sources, and j storage demands and the planning horizon consists of k time steps.
- Sources are represented by available storage reservoirs or EOR requirements for an oil field. Demands are comprised of fossil fuelled power plants where CC retrofit has been performed. The CO_2 captured from the plants will be transported to the reservoirs for storage/EOR usage. The CO_2 emission can only be stored in reservoirs which have been developed earlier. Should a power plant start to capture

Table 1 – Captured CO_2 load from power plants for hypothetical example.

Plants	Start to end (y)	Plant life considered in planning horizon (y)	Average captured CO_2 rate (Mt/y)	Total CO_2 load (Mt)
Plant A	0–30	30	10	300
Plant B	10–50	40	10	400
Total				700

CO_2 before a storage source is available, additional make-up storage sources (i.e. external service) is required to take the CO_2 load from the power plant.

- Since the sources (i.e. reservoirs) comprises of storage capacities which can generally store a large amount of CO_2, these storage sources can be linked to one or more demands (power plants) as the storage available can be shared among the plants.
- Due to the fact that the choice of pairing a CO_2 power plant to a designated reservoir is mainly driven by the project economics and geographical feasibility, the pairing of CO_2 emitting power plants to a particular CO_2 storage/EOR field will be justified only if a power plant linked to a reservoir will continue sending captured CO_2 load to the reservoir for the rest of the operating life of the plant, unless limited by its storage capacity and operating life.
- For a feasible planning problem, a reservoir is only made available to store captured CO_2 after it has been developed from time T onward. However, available storage capacity need not be used immediately upon becoming available. Conversely, plants with captured CO_2 will use up storage capacity immediately upon the generation of the CO_2 stream. Note that this asymmetric temporal relationship between the storage sites and power plants resembles the heat and mass transfer driving force in conventional pinch analysis, i.e. temperature in the case of heat recovery (Linnhoff et al., 1982), as well as concentration for mass integration and resource conservation (El-Halwagi, 2006; Foo, 2012). In other words, the sources (i.e., CO_2 storage sites) must be present before the demands (i.e., plants with captured CO_2).

3. Carbon storage composite curve

To handle the CCS planning problem illustrated above, a new graphical tool known as *carbon storage composite curves* (CSCCs) is proposed. As shown in Fig. 2, the CSCCs are plotted on a time vs. capacity diagram, where y-axis represents the time axis (usually in year) of the planning problem, i.e. storage and life span of power plants; while the x-axis represents the total accumulated CO_2 load (usually in Mt) involved for capture and storage.

For better illustration, a hypothetical example is used, with data in Table 1 represents the storage demand for plants A and B (with captured CO_2 load), both with different life duration and starting year of operation. For available storage, Reservoirs A and B will be developed to cater for their demands. The availability of the reservoirs varies, with Reservoir B developed 20 years after Reservoir A, as shown in Table 2.

As shown in Fig. 2(a), the storage reservoirs (storage sources) are first plotted as horizontal segments on the time versus capacity diagram, while CO_2 storage demands (from the power plants) are represented by diagonal segments. Note that the slopes of these segments represent the CO_2 flowrates for the given time frame. Both segments for the

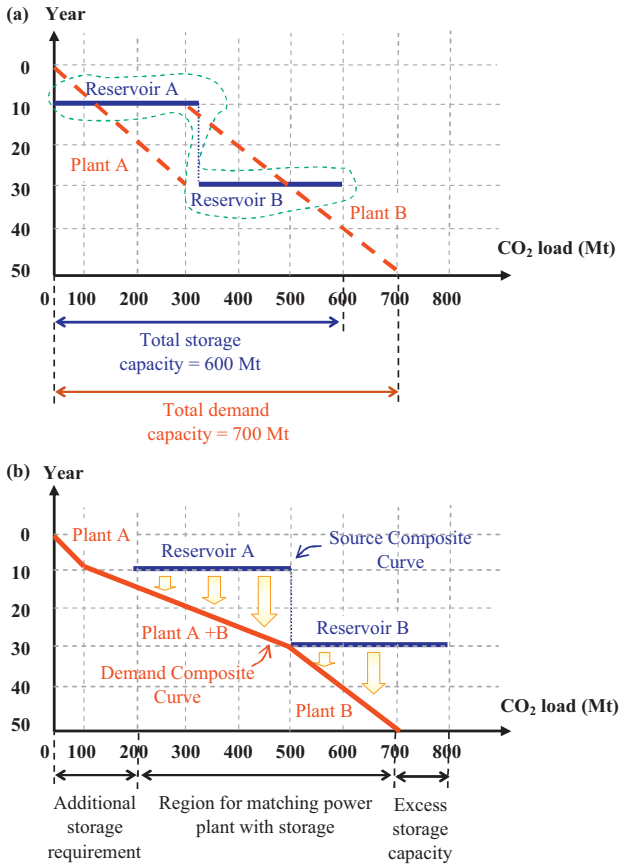


Fig. 2 – (a) CC with separate plot for storage reservoir and demands (power plants) for hypothetical case. (b) Combined CSCC for hypothetical example.

Table 2 – Storage capacity from reservoirs for hypothetical example.

Plants	Available from (y)	Total CO_2 load (Mt)
Reservoir A	10	300
Reservoir B	30	300
Total	–	600

storage sources and demands are then merged to become the *source* and *demand composite curves*. Note that the source composite curve takes the form of a "staircase", as it is constructed by various horizontal segments. In contrast, the demand composite curve is formed by merging the diagonal segments within each time interval. For instance, within the interval of 10–30 years where both plants A and B exist, their total CO_2 load are added, in order to form a single segment.

As mentioned earlier, for a feasible solution, the reservoirs (sources) are to be developed well ahead before they may be used for storing the capture CO_2 load. Hence, the source composite curve is moved horizontally to the right, until it is entirely above and just touching the demand composite curve

Table 3 – CO_2 storage capacity from reservoirs for case study.

Storage site	Start to end (y)	Reservoir life considered in planning horizon (y)	CO_2 injection rate (Mt/y)	Total CO_2 capacity (Mt)
Reservoir A	3–12	10	40	400
Reservoir B	8–18	11	60	660
Reservoir C	13–21	9	20	180
Total				1240

(left shift of the demand composite curve is also permissible; but not vertical shift). This corresponds to the case in Fig. 2(b). As shown, the arrows within the overlapping area between the two composite curves indicate the reservoirs (storage sources) are ready for use for the power plants (storage demands). In other words, 500 Mt CO_2 load can be sent from power plants to the storage reservoirs (in the reverse direction of the arrows), within the given time interval. The opening on the left side of the composite curves indicates that additional storage of 200 Mt is needed, due to the earliest storage by Reservoir A is only available from year 10 onwards. This additional storage requirement has to be fulfilled by outsourcing the storage with an external party (e.g. other nearby geographical region). On the other hand, the amount of excess storage capacity of the reservoirs is identified from the opening of the right side of the composite curve, i.e. 100 Mt from Reservoir B. This excess storage may then be used by other neighbouring geographical regions if they experience insufficient storage capacity problem. Note that the additional storage requirement and excess capacity in this problem are conceptually similar to heat integration problem where minimum hot and cold utilities are needed (Linnhoff et al., 1982); or the resource conservation networks with external resources and waste discharge (Foo, 2012). Finally, note that the point where the two composite curves touch is termed as the "*time pinch*", which is the bottleneck that limits the storage of CO_2 load from the power plants to the reservoirs.

From above illustration, it is clear that the CSCC provides valuable insights for CCS planning problem, particularly in identifying the bottleneck of the planning problem (i.e. the *pinch* point in the CSCC). Storage capacity deficiency can be mitigated from additional storage available from other reservoirs located sufficient close in geographical proximity. Excess storage capacity in the reservoirs can be addressed with allocation to other power plants in the region. A case study is next used to demonstrate the application of the CSCC.

4. Case study

For this case, the allocation problem will be addressed for a specific geographical region within a time planning horizon. Similar to the hypothetical case, the objective of this case study is to determine the amount of additional storage (if required) and also to identify excess storage capacity (if available). Table 3 shows the data of the reservoirs. As shown, Reservoirs A–C are identified for use, each with different storage capacity. Note that the total CO_2 storage capacity of the reservoirs is determined as 1240 Mt, with Reservoir A being the earliest available storage source, i.e. in year 3 (see Table 3).

Table 4 summarises the data for CO_2 load captured from power plants in consideration and to be sent for storage in the reservoirs. As shown, the total planning horizon for the case study is approximately 20 years, which is a reasonable period for energy planning. In practice, longer time horizons are subject to significant uncertainties. The power plant fleet comprises of coal, natural gas and oil power plants. To demonstrate the variation of power plants from the same fossil fuel type, two coal power plants A and B are considered in the planning horizon with coal power plant B assumed to be a new plant constructed as replacement for coal plant A. As such, coal power plant B requires storage for captured CO_2 towards the end of the planning horizon, i.e. from year 18 onwards. Note that the availability of coal power plant B is only 4 years although realistically plant operation life is more than 20 years. However, it is assumed that the plant is constructed late, and hence exceeds the planning horizon considered for this study. The remaining operating life of plant B can be considered in the next planning horizon.

CSCCs for the case study are plotted in Fig. 3. For clarity on the proposed methodology, the CSCC in Fig. 2 is further enhanced in detailed where the curves are plotted based on a year-by-year basis, such that the original diagonal demand composite curve also takes the form of a "staircase", as the CO_2 storage demand is now represented by horizontal bars, each equals the storage demand required from the power plants for a particular year. From Fig. 3, it is observed that there are potential matching regions for off-setting, identified from the overlapping region where both source and sink are available. The source composite curve is then shifted to the left until it touches the sink composite curve at the time pinch, i.e. year 8 (Fig. 4).

Table 4 – Captured CO_2 load from power plants for case study.

Power plant	Start to end (y)	Plant life considered in planning horizon (y)	Average captured CO_2 rate (Mt/y)	Total CO_2 load (Mt)
Coal power plant A	5–18	14	40	560
Natural gas power plant	6–15	10	20	200
Oil power plant	8–15	8	20	160
	16–17	2	40	80
Coal power plant B	18–19	2	40	80
	20–21	2	60	120
Total				1200

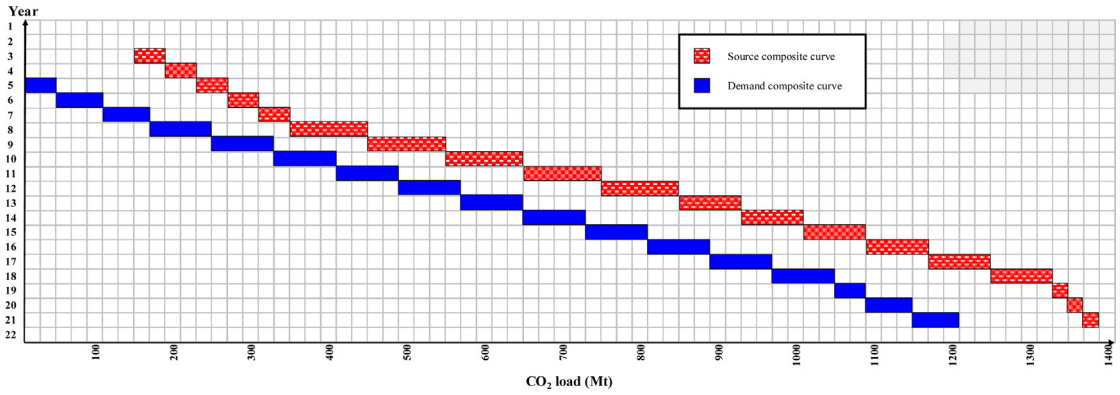


Fig. 3 – Plotting of CSCC for case study.

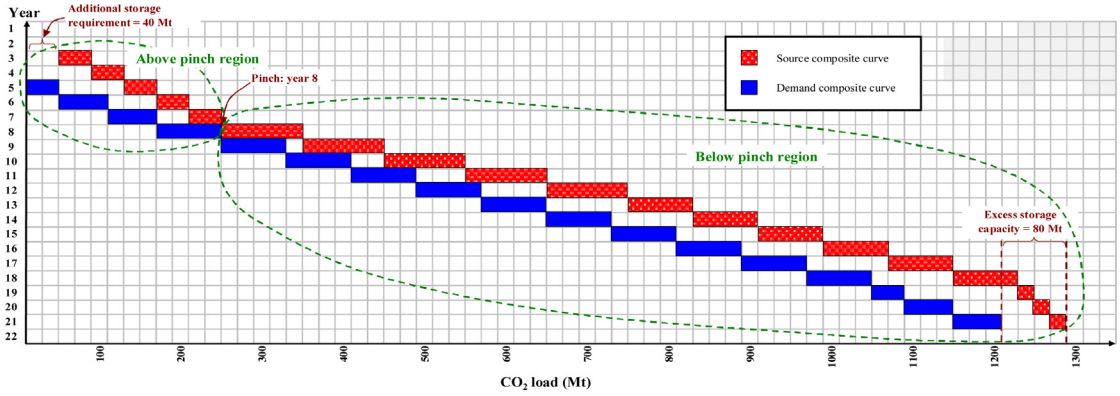


Fig. 4 – A feasible CSCC for case study.

We may also further categorise the region above and below pinch in Fig. 4. As shown, the amount of additional CO_2 storage requirement is identified from the region above the pinch as 40 Mt CO_2. Additional storage can be supplied from neighbouring storage sites in the area/region to support the storage deficit identified (i.e., excess CO_2 may be exported beyond the boundaries of the geographic region). In the region below the pinch, the amount of excess storage capacity from the reservoir is identified as 80 Mt. This spare capacity can be used to accommodate CO_2 storage from other additional power plants implemented with CCS initiatives in the area/region, or for the next planning horizon (i.e. as external storage available prior to the start of year 22). Having to identify the storage deficit enables the country to revise its energy plan so that external storage requirement can be reduced.

5. Cascade analysis for additional storage and excess capacity targeting

Apart from graphical approach, an algebraic technique is also developed in this work to address the same problem. This newly developed *carbon storage cascade analysis* (CSCA) is analogous to *material cascade analysis* reported for resource conservation networks (Manan et al., 2004; Foo et al., 2006; Foo, 2012). The general framework for CSCA is shown in Fig. 5. The case study is revisited and solved using CSCA to determine the minimum storage requirement for the captured CO_2 load. Similarly, the approach also determines the amount of excess CO_2 storage capacity available from the region.

Table 5 shows the basic framework of CSCA. The first step in conducting a CSCA is to arrange the demand and sources based on their time availability. As shown in the first two columns of Table 5, the availability time T_k is arranged in acceding order ($k = 1, 2, 3, \ldots, n$), with the earliest storage demand and source arranged at the top level. As explained earlier, CO_2 storage sources (reservoirs, with capacity F_{SRi}) on year T_k will only be available for storage from that time onwards and may not be used immediately once developed. For this example, an arbitrary step size (year by year) approach is applied to demonstrate the CSCA method, where the storage sources can cater to storage demand (captured CO_2 load from power plants, with capacity F_{SKj}) in the following year, i.e. T_{k+1}. For instance, Reservoir A is developed at time step $T = 3$, can be made available for storage use at time step $T = 4$. The demands and sources are summed at their respective time step and arranged in columns 3 and 4. Column 5 presents the

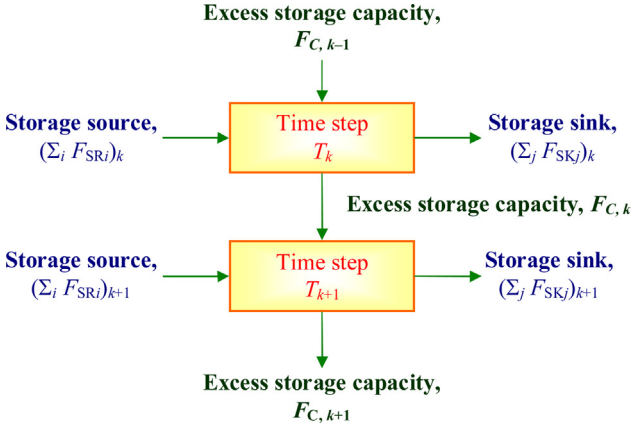


Fig. 5 – Concept of CSCA.

Table 5 – General framework for CSCA.

1 k	2 T_k	3 $\Sigma_j F_{SKj}$	4 $\Sigma_i F_{SRi}$	5 $\Sigma_i F_{SRi} - \Sigma_j F_{SKj}$	6 $F_{T,k}$
					F_{AD}
k	T_k	$(\Sigma_j F_{SKj})_k$	$(\Sigma_i F_{SRi})_k$	$(\Sigma_i F_{SRi} - \Sigma_j F_{SKj})_k$	
					$F_{C,k}$
$k+1$	T_{k+1}	$(\Sigma_j F_{SKj})_{k+1}$	$(\Sigma_i F_{SRi})_{k+1}$	$(\Sigma_i F_{SRi} - \Sigma_j F_{SKj})_{k+1}$	
					$F_{C,k+1}$
⋮	⋮	⋮	⋮	⋮	⋮
$n-2$	T_{n-2}	$(\Sigma_j F_{SKj})_{n-2}$	$(\Sigma_i F_{SRi})_{n-2}$	$(\Sigma_i F_{SRi} - \Sigma_j F_{SKj})_{n-2}$	
					$F_{C,n-2}$
$n-1$	T_{n-1}	$(\Sigma_j F_{SKj})_{n-1}$	$(\Sigma_i F_{SRi})_{n-1}$	$(\Sigma_i F_{SRi} - \Sigma_j F_{SKj})_{n-1}$	
					$F_{C,n-1} = F_{EX}$
n	T_n				

net capacity $\Sigma_i F_{SRi} - \Sigma_j F_{SKj}$ between the storage sources and demands at the time point k; with positive indicating surplus and negative indicating deficit.

Next, the net CO_2 flowrate surplus/deficit is cascaded down the time intervals to yield the cumulative surplus/deficit capacity ($F_{C,k}$) in column 6. The largest deficit value in this column represents the minimum required storage capacity for a feasible cascade solution. The cascade exercise is carried out again with the identified storage capacity requirement (F_{AD}). Excess storage capacity (F_{EX}) of the CCS network is determined from the last time interval in the feasible cascade table (where no deficit value found in column 6).

Table 6 – CSCA for case study.

k	T_k (y)	$(\Sigma_i F_{SRi})_k$ (Mt/y)	$(\Sigma_j F_{SKj})_k$ (Mt/y)	$(\Sigma_i F_{SRi} - \Sigma_j F_{SKj})_k$ (Mt/y)	$F_{C,k}$ (Mt/y)
					$F_{AD} = 40$
1	4	40		40	
					80
2	5	40	40	0	
					80
3	6	40	60	−20	
					60
4	7	40	60	−20	
					40
5	8	40	80	−40	
					0
6	9	100	80	20	(Pinch)
					20
7	10	100	80	20	
					40
8	11	100	80	20	
					60
9	12	100	80	20	
					80
10	13	100	80	20	
					100
11	14	80	80	0	
					100
12	15	80	80	0	
					100
13	16	80	80	0	
					100
14	17	80	80	0	
					100
15	18	80	80	0	
					100
16	19	80	40	40	
					140
17	20	20	60	−40	
					100
18	21	20	60	−40	
					60
19	22	20	0	20	
					$F_{EX} = 80$

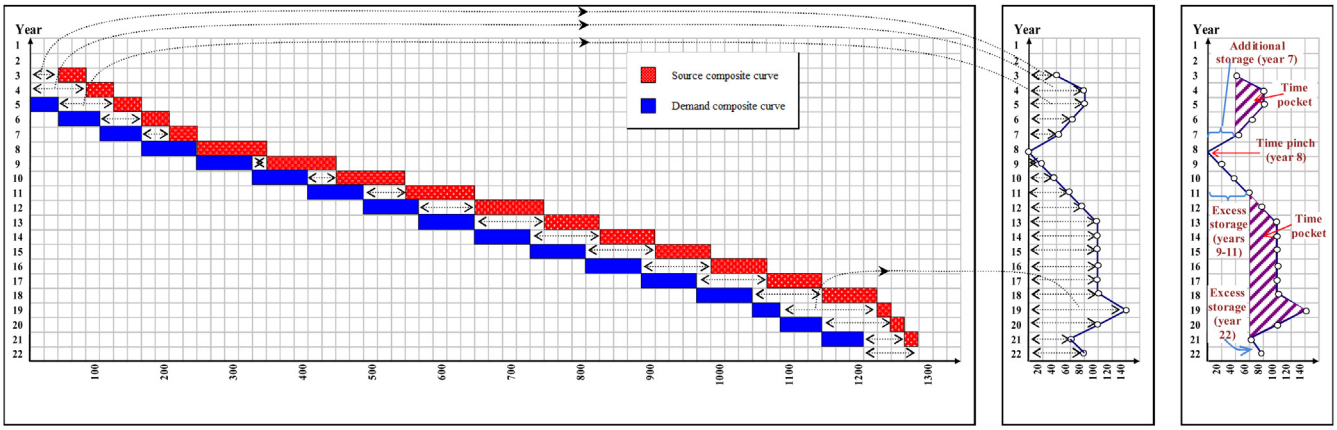


Fig. 6 – Time grand composite curve (GCC) plotted graphically to illustrate horizontal gaps between sink–source composite curve.

The case study is revisited with CSCA, with results shown in Table 6. The amount of additional storage capacity requirement (F_{AD}) is determined as 40 Mt CO_2, while the total excess storage available for other use (F_{EX}) is determined as 80 Mt CO_2 from the last entry of column 6. Both values are consistent with the results reported by CSCC in Fig. 4.

Note that the CSCA provides quantitative results which are more accurate as compared to the graphical method using the CSCC the graphical method provides conceptual insights on the solution as rough estimation on the CO_2 load quantity can be deciphered from the CSCC. On the other hand, cascade analysis is a numerical tool which is preferred when rapid and accurate answers, or when repeated calculation is required. This technique for example is easily implemented in the form of a spreadsheet. The results from Table 6 can also be applied in plotting the *grand composite curve*, which will be discussed in the following section.

6. Scheduling for additional and excess storage

The feasible CSCC in Fig. 4 determines the additional storage requirement and excess capacity of the CCS planning problem. However, these requirements may not be necessarily made available at the start time of the planning horizon, which allows time for resources to be deployed for activities such as conducting additional geological surveys of candidate storage sites, etc. As indicated in the CSCC in Fig. 4, additional storage requirement of 40 Mt is required in year 3. This does not reflect the planning scenario where the earliest CO_2 load captured from power plants for storage only starts in year 5 (see Table 4). As such, the actual timing of the additional and excess storage can be determined from the *grand composite curve* (GCC), as illustrated in Fig. 6. The GCC curve is plotted by transferring the horizontal differences of the feasible CSCC from Fig. 4 for every year of the planning horizon.

From the GCC in Fig. 6(c), *time pockets* (represented by the shaded areas) are first identified. This corresponds to the heat pocket in thermal pinch analysis, in which process-to-process heat transfer occurs without the use of external hot or cold utilities (Linnhoff et al., 1982). In other words, available heat source (to be cooled) complements the required heat sink (to be heated). For this problem, the time pockets represent the time interval where the available CO_2 storage capacities from the sources can accommodate the storage requirement (i.e. CO_2 load captured from the power plants). This also represents the corresponding time interval or duration where the sink complements the sources without additional make-up requirements. The open area above the time pinch in Fig. 6(c) indicates that the additional storage of 40 Mt CO_2 storage requirement is only needed in year 7. Hence, it is noticed that the GCC provides accurate representation on the timing of the additional storage requirement. The open area below the time

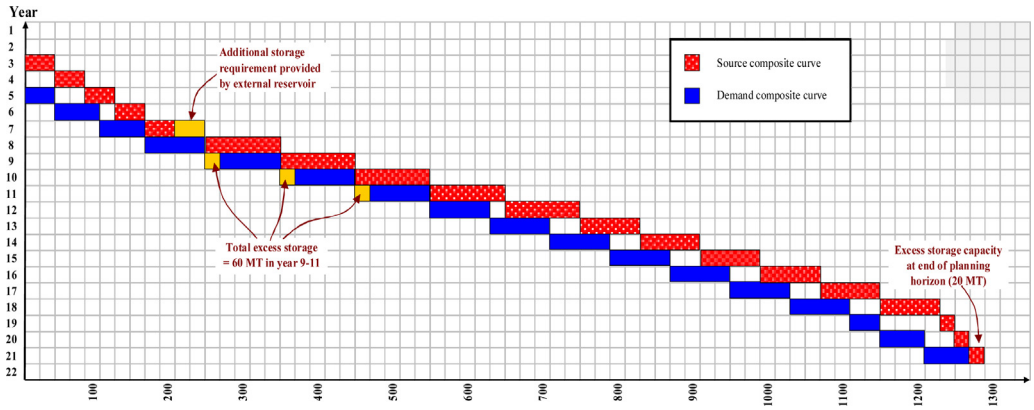


Fig. 7 – Detailed CSCC for Option 1 of sink-source pairing.

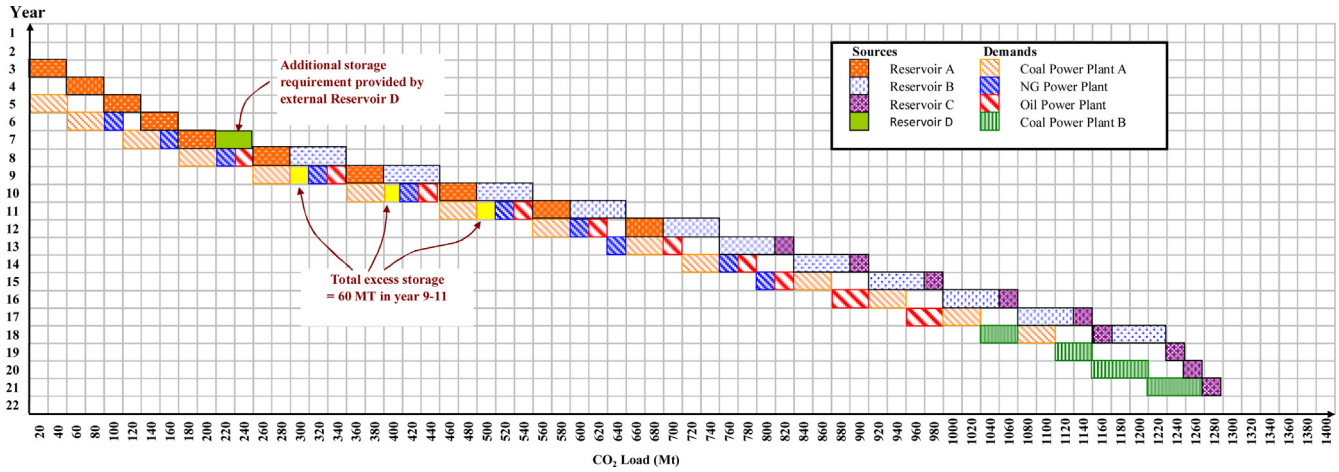


Fig. 8 – Detailed time composite curve for Option 1.

pinch represents the years and corresponding amount where excess storage is available from the storage reservoirs. From Fig. 6(c), cumulative excess storage from year 9 to year 11 totals to 60 Mt CO_2, while that for year 22 corresponds to 20 Mt CO_2, leading to the total of 80 Mt CO_2. This result is consistent with the initial estimation of excess storage capacity represented by the sink–source composite curves in Fig. 4.

Results from the GCC in Fig. 6 may also be transferred to form the "balanced CSCC" given in Fig. 7, to include details on the actual timing of the additional and excess storage available in the planning horizon. The segments of the source composite curve from year 3 to year 7 are shifted to the left of the time pinch (i.e. year 8). The empty slot (where the composite curve detaches from the original curve) in year 7 now confirms the requirement of 40 Mt CO_2 storage identified by the GCC. Similarly, segments of the demand composite curve from year 9 to year 21 are shifted to the right of the time pinch. Three (3) empty slots are created from the shift now represents the amount of excess storage available at 20 Mt CO_2 per year from year 9 to year 11, as total storage capacity of 100 Mt CO_2 exceeds the required storage demand from the power plants of 80 Mt CO_2. Note also there exist an excess storage capacity of 20 Mt at the end of the planning horizon (in which can only be utilised in the next planning horizon), making the total excess capacity of 80 Mt.

7. Detailed allocation and matching of CO_2 sources with storage reservoir

Following the identification of planning targets, the case study is revisited to determine the allocation of storage demands and sources. In Sections 4–6, the planning problem is solved using a macroscopic approach, without exact allocation of CO_2 load from power plants to reservoirs.

The CSCC in Fig. 7 is further refined by breaking down the demand and source composite curves into their individual constituents (i.e., power plants of coal, natural gas and oil; Reservoirs A–C), instead of plotting for total storage capacity and requirement. To determine a possible pairing combination, the individual source and demand for each time interval will be rearranged to ensure that once a power plant is paired to a reservoir, the plant will continue sending captured CO_2 load to the same reservoir unless limited by the reservoir storage capacity, or operating life, as explained in Section 2. This will ensure an economically viable pairing option to be identified, as setting up CCS facilities between a plant and storage site often entails costly facilities (e.g. pipeline and booster compressors). As such, limiting the number of routes from a plant to a reservoir (where possible) will be the primary approach adopted when determining possible permutation options in pairing.

For the case study above, two (2) possible allocation options have been identified using the detailed CSCC, which is differentiated by the variation in pairing a power plant to a reservoir for CO_2 storage. Detailed CSCC for Option 1 is shown in Fig. 8. The allocation and connectivity between power plant and reservoir is better explained with the schematic representation in Fig. 9. A total of 10 connections are established between the power plants and reservoirs. The spare storage capacity available from year 9 to year 11 is derived from Reservoir B to support neighbouring plant requirements in the area/region. Detailed allocation and breakdown on the CO_2 flowrate transported from the power plants to the reservoir is summarised in Table 7 for easy reference. Based on results in Option 1, CO_2 captured from coal power plant A will be sent to Reservoir A

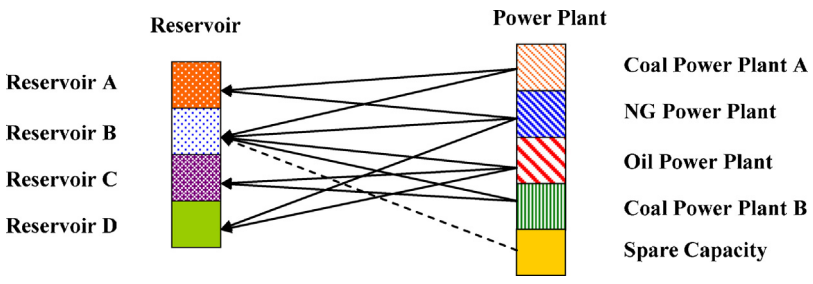


Fig. 9 – Schematic representation of sink–source pairing for Option 1.

Table 7 – Detailed allocation between captured CO_2 in power plant and reservoirs (Option 1).

From	To	Duration (y)	CO_2 flowrate (Mt/y)
Coal power plant A	Reservoir A	5–13	40
	Reservoir B	14–18	40
Natural gas power plant	Reservoir A	6–7	20
	Reservoir D	8	20
	Reservoir B	9–15	20
Oil power plant	Reservoir D	8	20
	Reservoir B	9–14	20
		16–17	20
	Reservoir C	15–17	20
Coal power plant B	Reservoir B	18–19	20
		20–21	60
	Reservoir C	18–19	20
		21	60

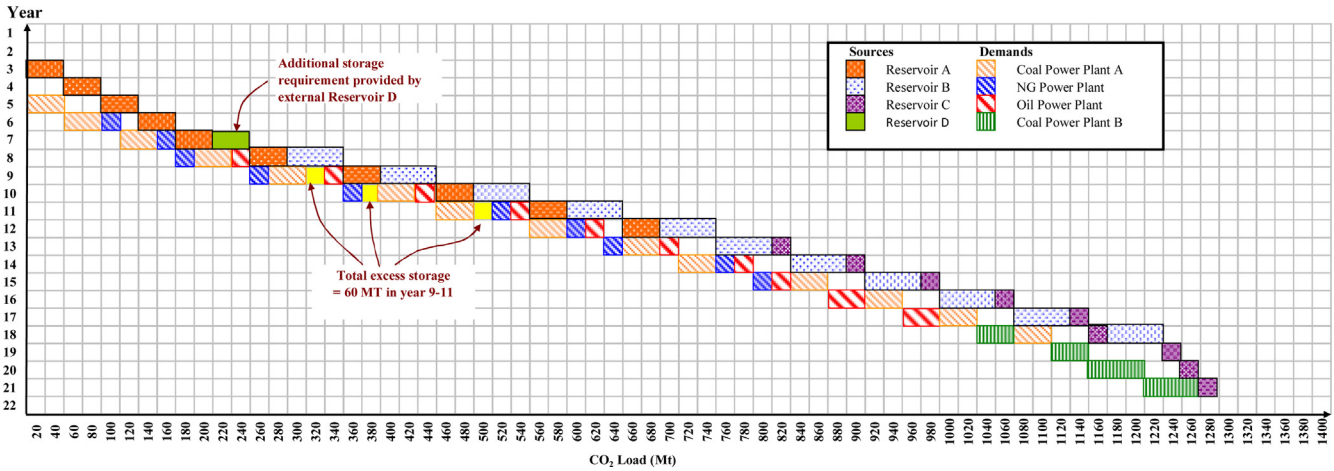


Fig. 10 – Detailed CSCC for Option 2 of sink-source pairing.

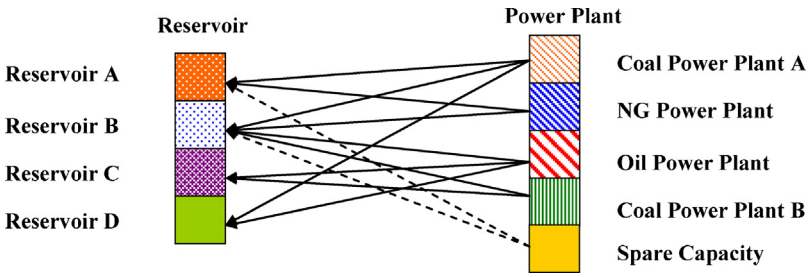


Fig. 11 – Schematic representation of sink–source pairing for Option 2.

for storage from year 5 to year 13 until the end life of Reservoir A after which alternative Reservoir B is required for storage of remaining captured CO_2 from coal power plant A (each with CO_2 flowrate of 40 Mt/yr).

Captured CO_2 from natural gas power plant will be sent to Reservoirs A, B and D (additional make-up storage) throughout the operating life of the plant. Also, captured CO_2 from oil power plant will be sent for storage in Reservoirs B–D from year 8 to year 17.

Captured CO_2 from coal power plant B will be sent to Reservoirs B and C when it starts operation in year 18. Since the planning horizon considered in this study is only 20 years, the storage strategy for coal power plant B from year 22 onwards will be considered in next planning horizon (assuming that coal power plant B, Reservoirs B and C are still available in the new planning horizon).

Spare storage capacity has been identified for Reservoir B from year 9 to 11 to accommodate additional captured CO_2 from the power plants or new plants located in the region.

Option 2 provides another possible solution for the allocation problem and is represented in Figs. 10 and 11. In addition, in this option, a total of 10 connections are established between the power plants and reservoirs. Detailed allocation and breakdown on the CO_2 flowrate transported from the power plants to the reservoir is tabled in Table 8 for easy reference.

Table 8 – Detailed allocation between captured CO_2 in of power plant and reservoirs (Option 2).

From	To	Duration (y)	CO_2 flowrate (Mt/y)
Coal power plant A	Reservoir A	5–7	40
		8–9	20
		11–13	40
	Reservoir B	9	20
		14–18	40
Natural gas power plant	Reservoir A	6–7	20
	Reservoir D	8	20
	Reservoir B	9–15	20
Oil power plant	Reservoir D	8	20
	Reservoir B	9–14	20
		16–17	20
	Reservoir C	15–17	20
Coal power plant B	Reservoir B	18–19	20
		20–21	60
	Reservoir C	18–19	20
		21	60

For Option 2, spare storage capacity has been identified for Reservoir A for year 10 and Reservoir B for year 9 and year 11 to accommodate additional captured CO_2 from the power plants or new plants located in the region.

8. Conclusion

A pinch-based graphical tool known as *carbon storage composite curves* (CSCC) has been developed for use in the selection and allocation of CO_2 storage capacity with power plants that implement carbon capture (CC). The proposed graphical approach with CSCC serves as a valuable decision support tool for providing planning insights with CCS deployment in a geographic region under temporal constraints. In addition, a grand composite curve (GCC) has been developed for scheduling of storage capacity surplus or deficit. The required additional storage capacity can be identified quantitatively and planned for remedial back-up in the required years to ensure that adequate CO_2 storage support is provided. Similarly, amount of excess CO_2 storage is identified together with the time of availability. This will help to provide insights on the spare capacity available in the area/region which can be utilised by other power plants that may be considered for CO_2 capture later, or which may be utilised by neighbouring geographical regions, or in the new planning horizon.

The carbon storage cascade analysis (CSCA) method can then be applied to provide more precise algebraic computations that are equivalent to CSCC. Detailed mapping on the allocation can also be executed using the detailed CSCC. Detailed CSCC identifies possible options as roadmaps to be adopted during preliminary planning and feasibility assessment in constructing the CCS network. These options do not consider the economic aspects of infrastructure investment as such issues are better addressed using detailed optimisation models during subsequent stages of planning. Based on the options identified, further assessment shall be conducted with reference to timing, resources, project economics and environment factor before a roadmap is selected for execution.

For future work, multi-region systems can be used to optimise CO_2 management planning and storage strategy implemented for multiple regions around the world and not limited to a single region. In other words, the captured CO_2 can be sent for storage in reservoirs located in another region while reservoirs can be utilised to accommodate CO_2 storage for power plants in other regions. This is similar to previous work on inter-plant water integration by Chew and Foo, 2009. In addition, the graphical approach may be improved to account for injection rate limits of storage sites.

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